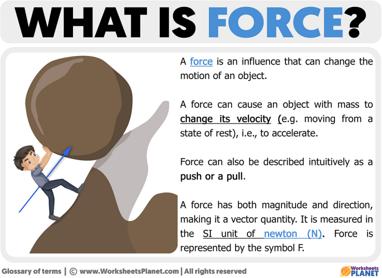
Definition

A force is a push or a pull that acts upon an object as a result of its interaction with another object. Forces result from interactions and can affect how an object moves. They may cause an object to start moving, stop moving, or change direction. The description of a force would be incomplete if direction is not given, as it's a vector quantity, meaning it has both magnitude (how strong it is) and direction.
Let's take some real-world examples to understand this concept better. Think about playing soccer. When you kick the ball, you are applying a force to it. This force, depending on how hard you kick the ball, will cause it to move in a certain direction. Similarly, if you've ever tried to push a heavy object like a piece of furniture, you've experienced force. The harder you push (the greater the force), the more the object moves.
But remember, force is not just about moving objects. It's also about stopping them. Imagine a car coming to a stop. It's the force applied by the brakes that reduces the speed and eventually stops the car.
Now, can you think of any other examples of force in your everyday life? And do you remember what we mean when we say force is a 'vector'

Explanation with Example

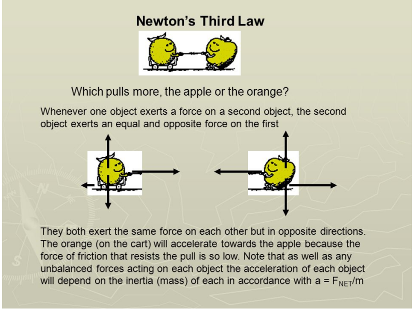
- Force is a push or a pull acting upon an object.
- It results from the object's interactions with another object.
- Force can cause an object to start or stop moving, or to change direction.
- Force is a vector quantity, meaning it has both magnitude (strength) and direction.
- Real-world examples of force include kicking a soccer ball or pushing a piece of furniture.
- Force is not just about moving objects, but also about stopping them. For instance, a car comes to a stop due to the force applied by the brakes.

Image 1



What is Force — Definition & Examples

Image 2



Newton's Third Law Whenever one object exerts a force on a second ...

Follow-Up Questions

- Can you think of any other examples of force in your everyday life?
- What do we mean when we say force is a 'vector' quantity?
- How does the concept of force apply to other activities you engage in?
- Can you think of ways in which understanding force can be useful in your everyday life?
- What other questions do you have about the concept of force?

References

- https://www.physicsclassroom.com/class/newtlaws/lesson-2/the-meaning-of-force
- https://www.reddit.com/r/askscience/comments/15meoc2/if_newtons_third_law_says_that_if_you_exert_a/
- https://www.physicsclassroom.com/class/newtlaws/lesson-4/newton-s-third-law